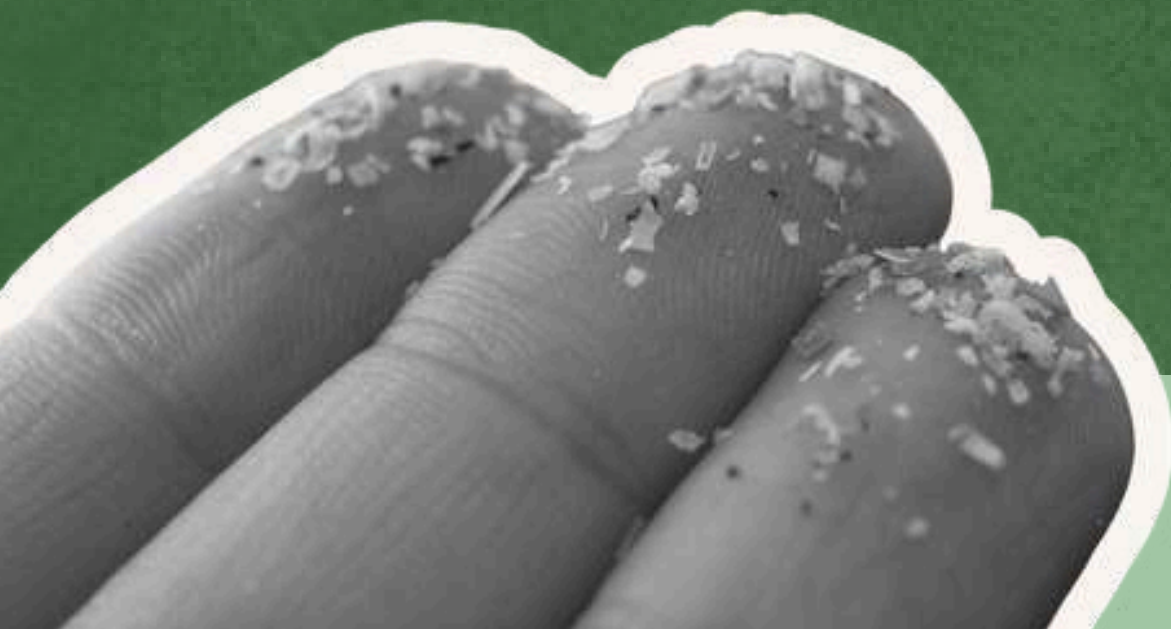
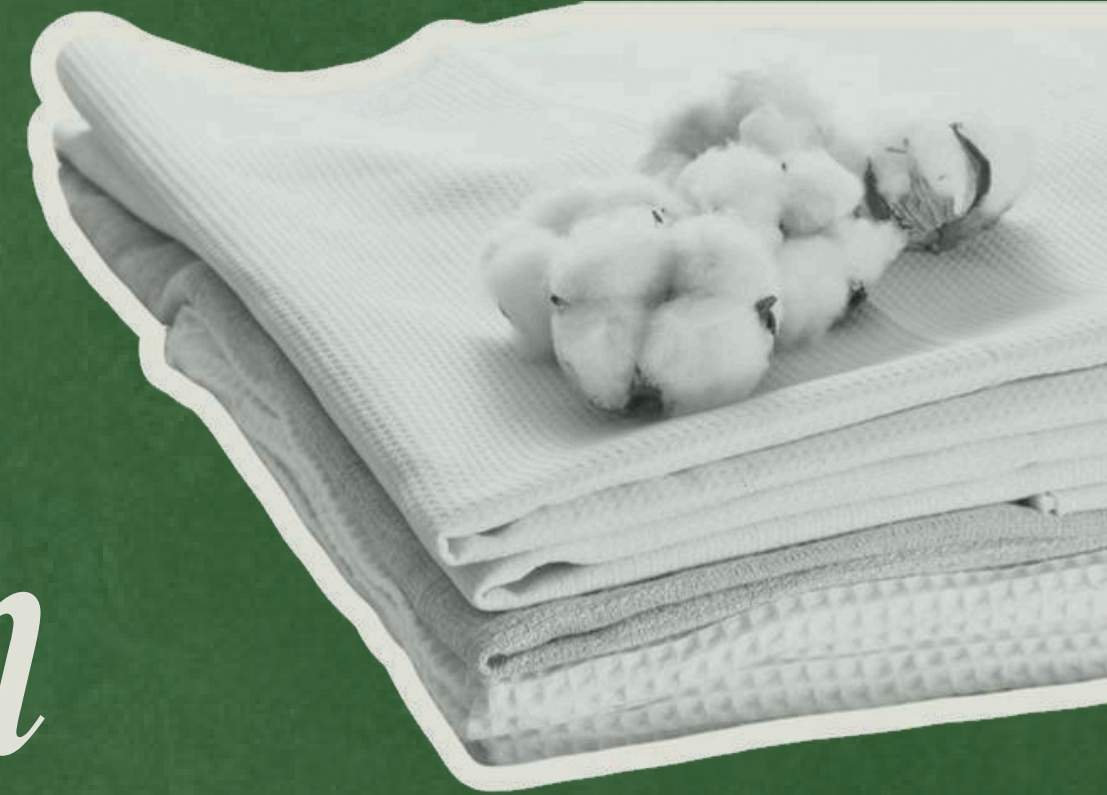




Microplastics in Swiss Textiles

Let's work for a Better Planet



Team 1: Hanieh Jebeli, Rana Okutan, Divya Shori



Why This Matters

Microplastic pollution isn't an abstract statistic it's already inside us.



Found in human blood

Microplastics have been detected in human blood, lungs, and placental tissue evidence our bodies are already exposed.



Ingested by marine life

Documented in the digestive tracts of hundreds of marine species, from plankton to whales entering the food chain we depend on.

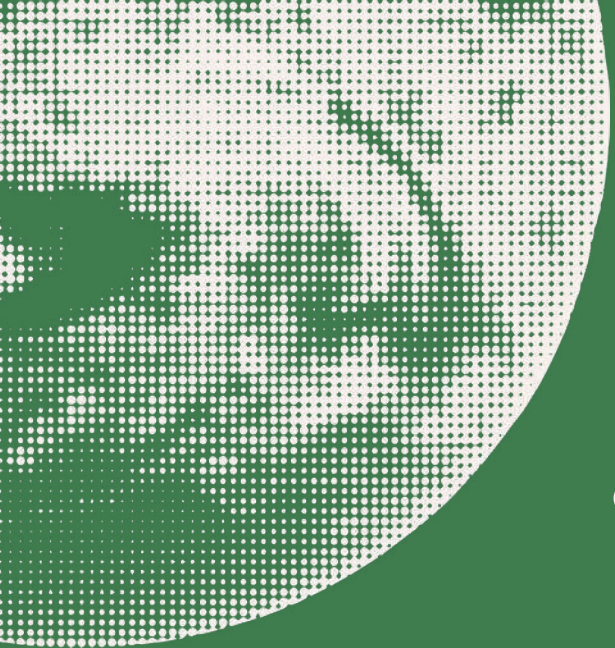


Effectively permanent

Synthetic fibres don't biodegrade. They fragment into smaller and smaller pieces that can persist in soil and water for centuries.

measuring Switzerland's contribution — identifying what's avoidable — matters beyond the numbers.

Sources: Leslie et al., Environment International, 2022 (microplastics in human blood) — Ragusa et al., Environment International, 2021 (placenta) — GESAMP (2015), marine species ingestion reviews



Fashion has quietly become a plastics industry

69%

of global fibre production
is now synthetic

59%

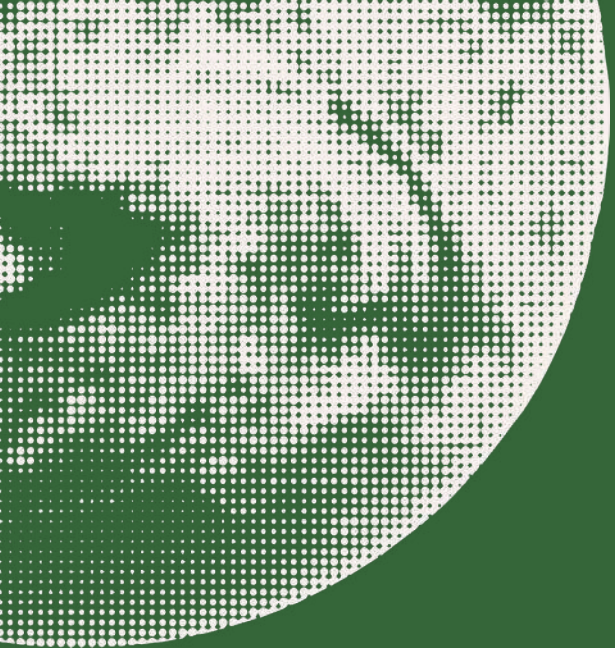
of all fibre worldwide
is polyester alone

40×

more plastic reaches Swiss
soil than freshwater

Switzerland has very limited domestic textile production today, meaning its textile footprint is largely shaped by the products it imports and consumes.

Sources: Textile Exchange, Materials Market Report (69% synthetic, 59% polyester) — Kawecki & Nowack, Env. Sci. Technol. 53(16), 2019 (40× soil vs freshwater)



Target

Quantify Switzerland's textile driven microplastic footprint

1

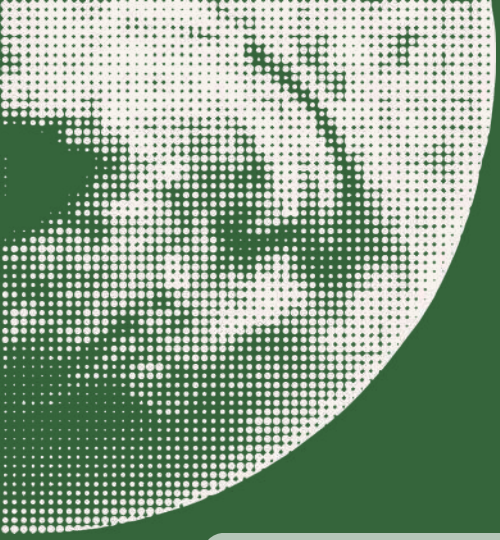
How much textile-derived
microplastic pollution does
Switzerland generate?

2

How much of that pollution can we
confidently quantify?



Data Sourcing



Data search: How we found data sources

1

SwissImpex / BAZG

Official Swiss customs data, identified via opendata.swiss as the source for textile & apparel import volumes.

2

Kawecki & Nowack, 2019 (Empa)

Found by searching the exact seven-polymer structure of our EMPA file.

3

De Falco et al., 2021

Literature search for peer-reviewed fibre shedding rate studies.

4

Conley et al. / Kaegi et al.

Literature search for wastewater treatment plant microfibre removal studies.

Data Integration in Practice

```
library(readr)
library(dplyr)

data <- read_csv("textile_apparel_imports.csv")
synthetic_share <- 0.69 # Textile Exchange

yearly <- data %>%
  group_by(year) %>%
  summarise(
    total_imported_kg = sum(Total_quantity_kg, na.rm = TRUE)
  ) %>%
  mutate(
    estimated_synthetic_kg = total_imported_kg *
synthetic_share
  )

write_csv(yearly, "textile_imports_by_year.csv")
```

1

Reads the 791,000-row file already filtered to CPA13/14 (textiles & apparel)

2

Groups every import record by year and sums total kg.

3

Applies the 69% global synthetic-fibre share to estimate the synthetic portion

4

Writes one clean row per year

From raw customs data to model input

A transparent pipeline showing what is observed Swiss data and what is a literature-based assumption.





Structural Model

How synthetic clothing becomes microplastic pollution

A simple pathway from imported textiles to environmental release.



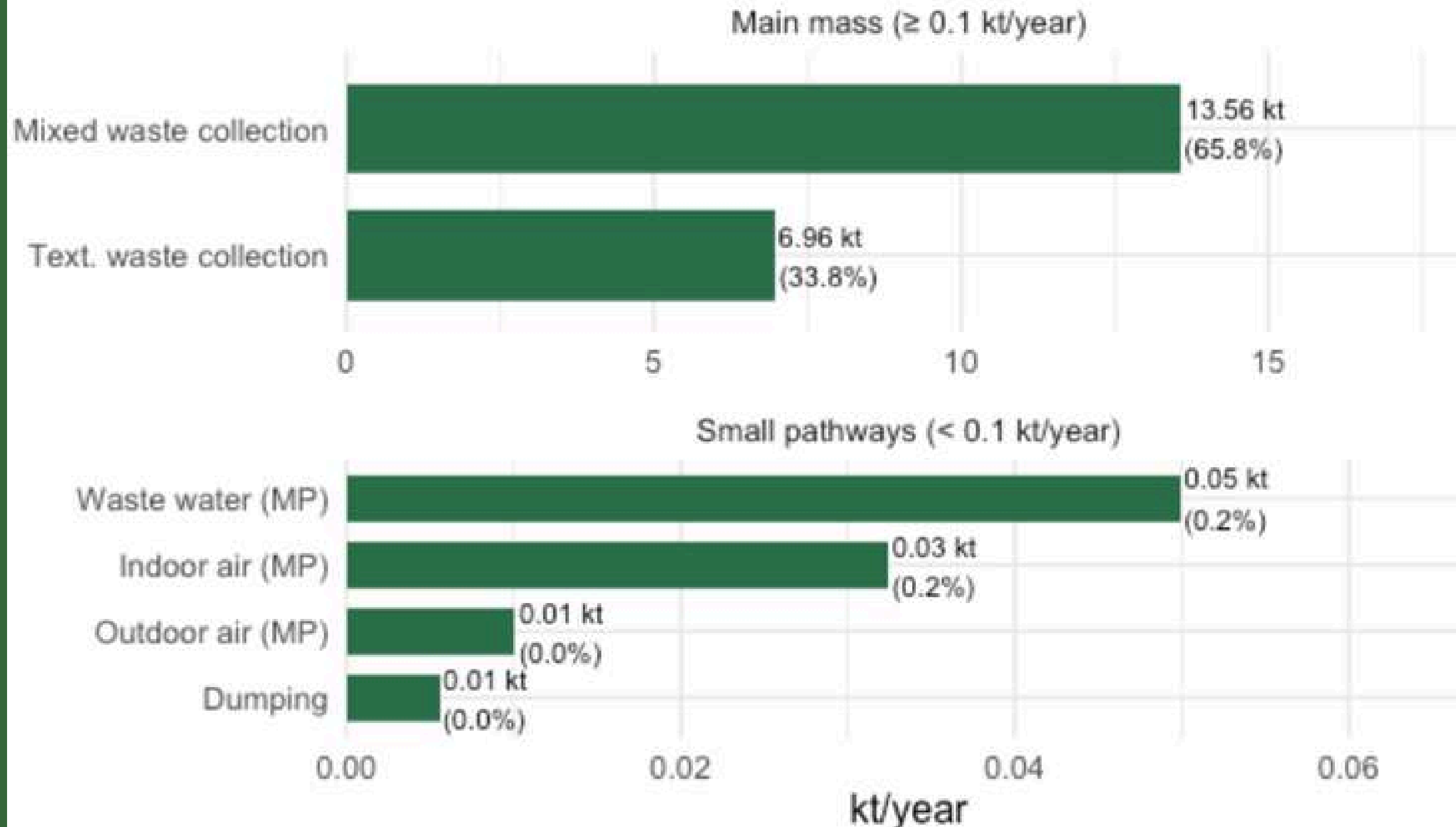
Model levers: textile imports × shedding rate × (1 – WWTP capture rate) → microfibres released

System View

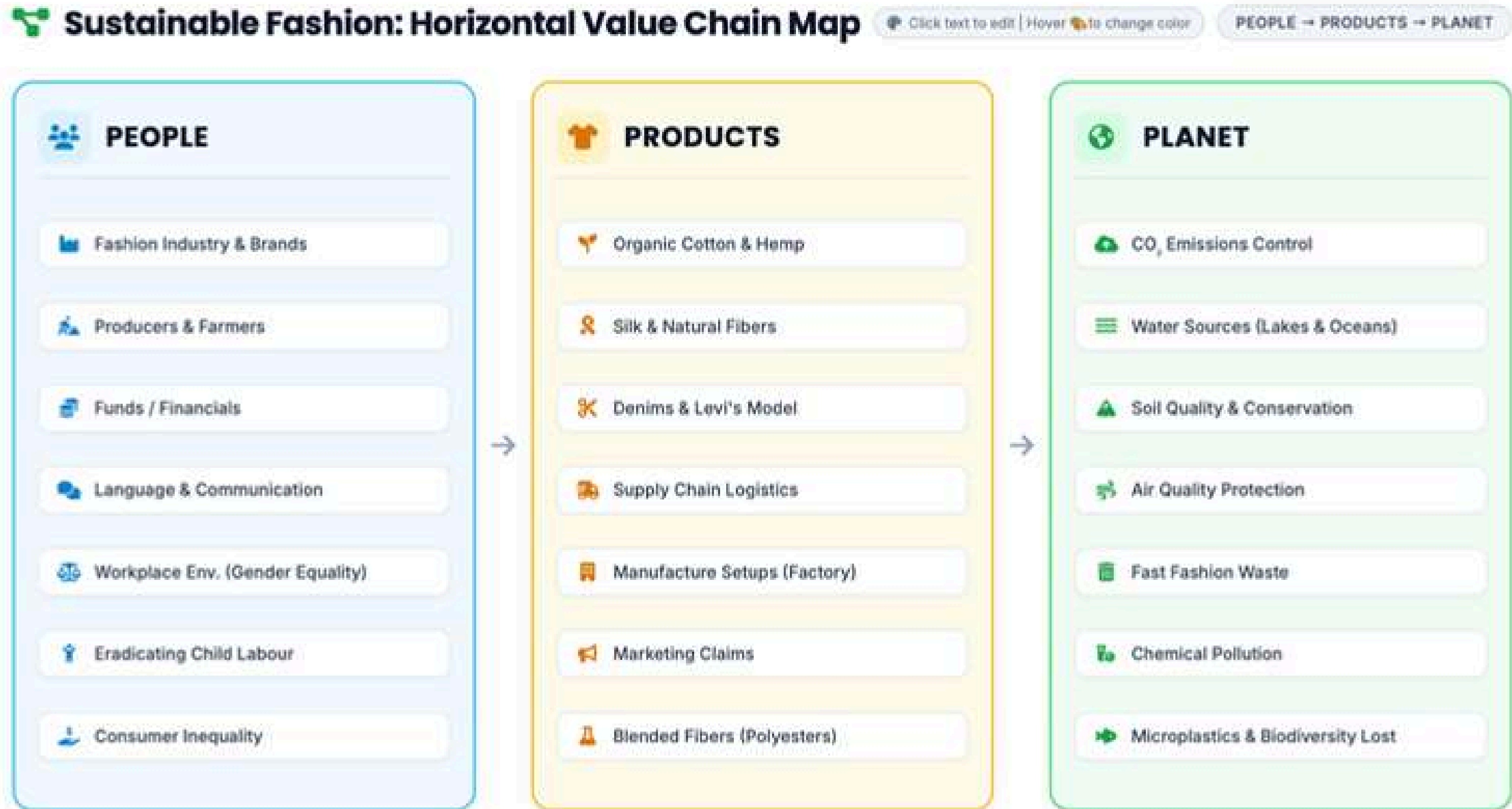
Sustainability Analytics

PET clothing: first destinations after use

Unit: kt/year — panels use different x-scales on purpose



System View

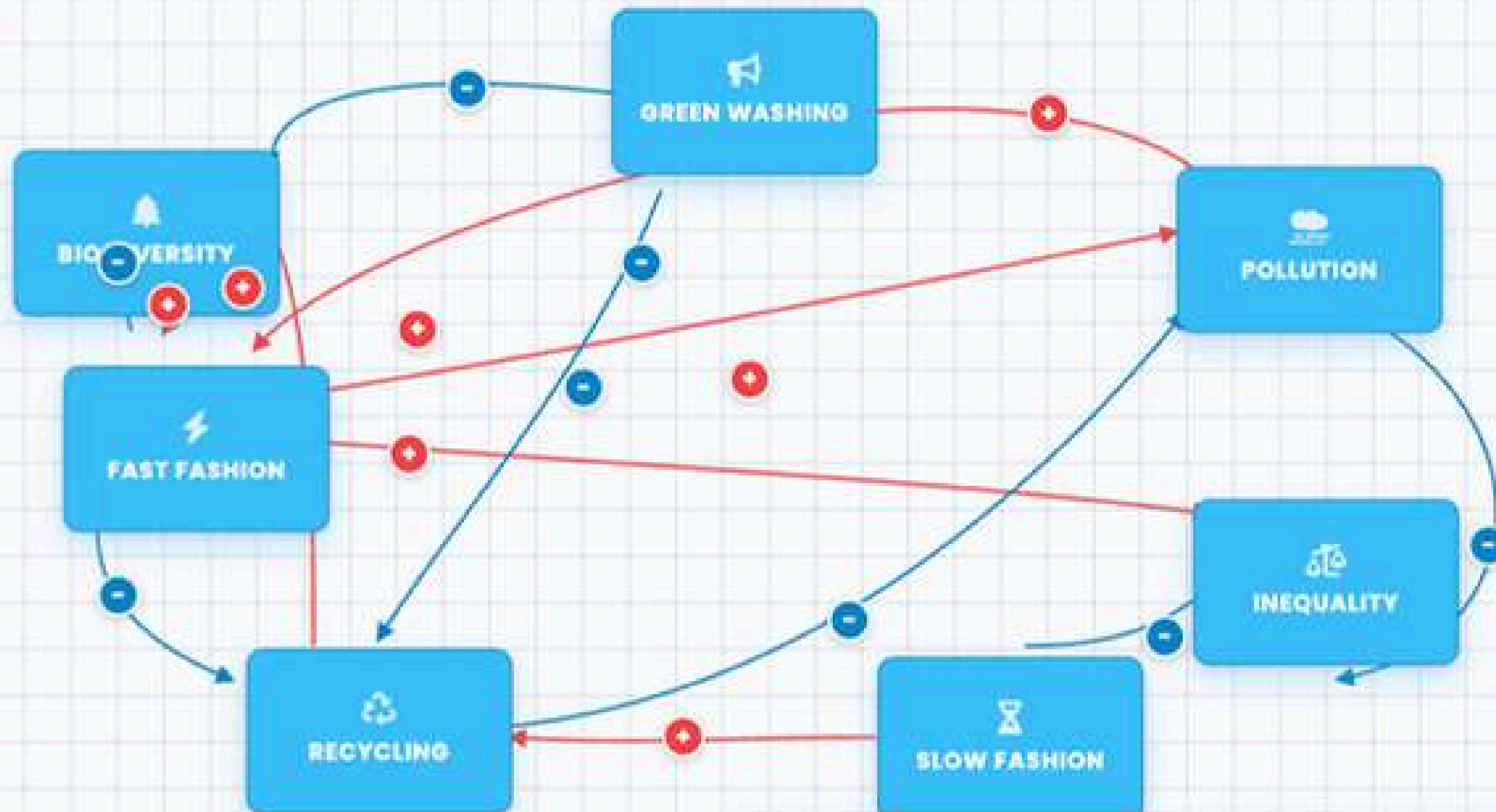


Ecosystem System Map



Systemic Feedback Loop & Causal Dynamics

EXACT DIGITIZED CAUSAL LOOP DIAGRAM



 Reinforcing Feedback Loop (Increases Effect)  Balancing Feedback Loop (Decreases / Dampens Effect)

System Dynamics: Recycling Simulation & Micro-Trends

SIMULATION & DATA MODEL

SIMULATE HYPOTHESIS

What happens if we do more recycling?
Over time the predictors can change → We get the microscopes →
System Dynamics modeling.



Data Inputs:

- 1 Microplastics Metric
- 2 Import Textiles Volumes

 Biodiversity (Textile)

 Microplastics
trend in Swiss
textile

 Microplastics

Why / How Dynamics

 Fast Fashion Textiles

 Slow-Fashion Textile

- ✓ Long Run Use Model
- ✓ Natural & Eco-Friendly Fibers

 Microplastic (Pollution)

How?

Presentation →

GOAL ?

FUTURE →

Ecological Loop Analogy

Fox

Rabbit

+ Feedback

- Feedback

How?

Sustainability Background

The Global Goals as our Benchmark



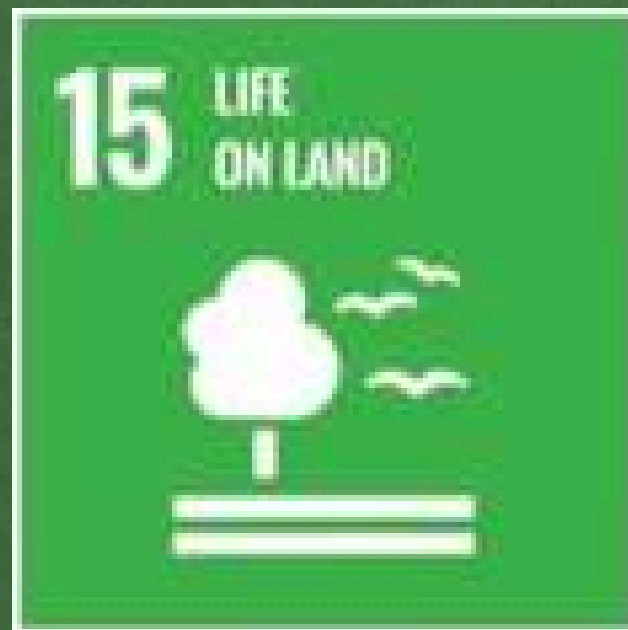
The core: synthetic clothing is made and discarded faster than the planet absorbs its microfibre.



Fibres shed in the wash slip through wastewater treatment into rivers and lakes.



Synthetic textiles are the #1 source of primary ocean microplastics (IUCN ~35%).



Landlocked Switzerland's fibre routes to farmland soil via sewage sludge.



Our response: cleaner fibres, capture tech and the Agentic AI verification agent.



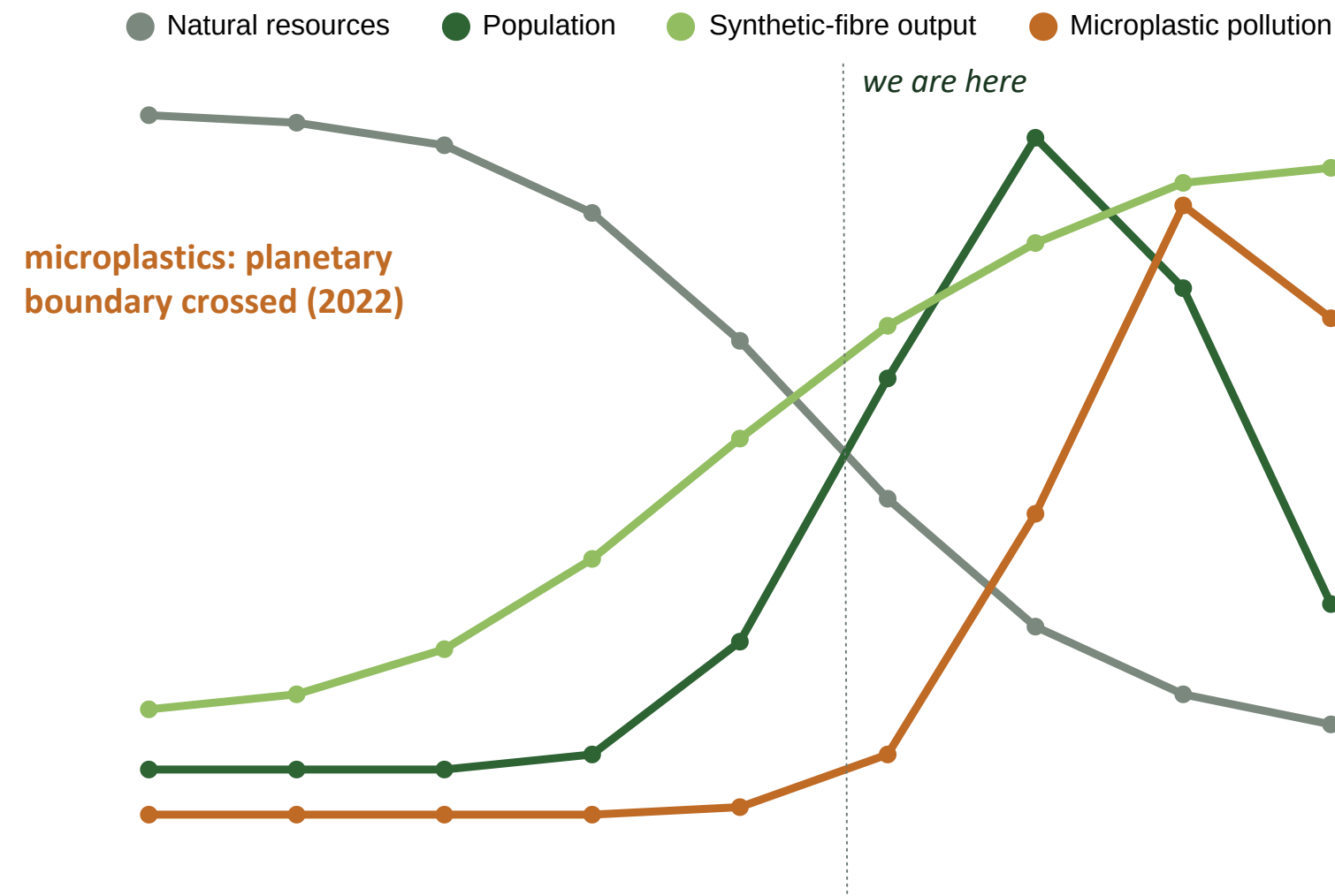
Polyester is fossil-based, 57% of all fibre and rising, a growing carbon load.



Fashion is a systems problem

Synthetic-fibre output is still on the rising limb; microplastic pollution is the accumulating overshoot

Fashion on the limits-to-growth curve



Limits to Growth

Club of Rome, 1972 · 50-year update

World3 → iSDG

Forrester → Millennium Institute system models

Planetary Boundaries

novel entities (plastics) crossed, 2022

WEF Global Risks

pollution & Earth-system risks, top-ranked

Doughnut Economics

overshoot of the ecological ceiling

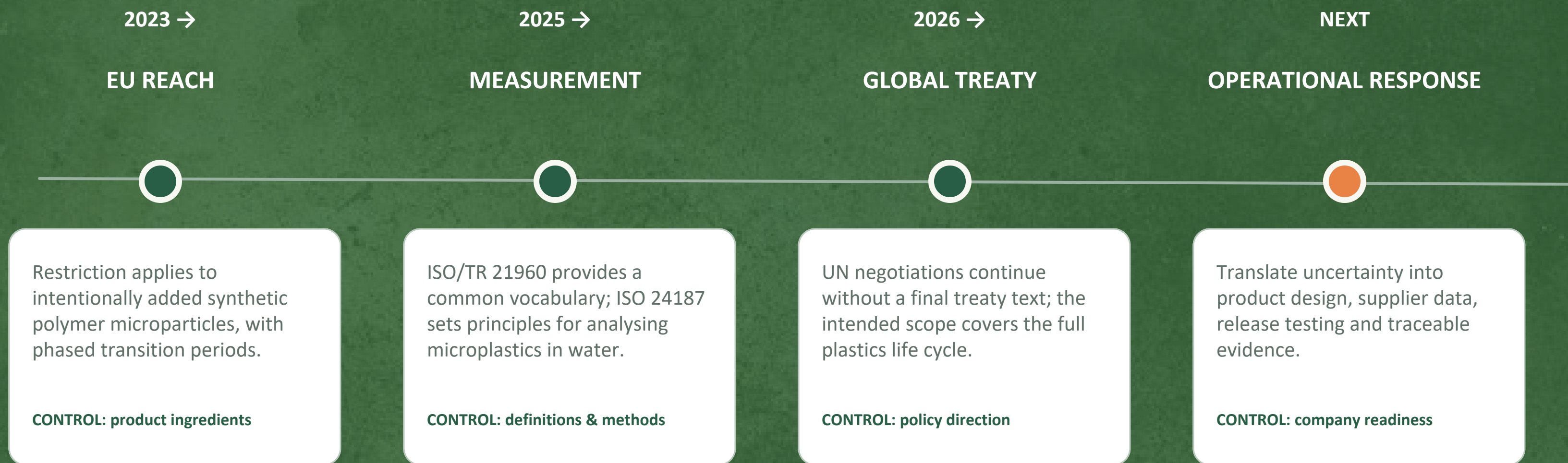
→ So we modelled microfibre as a system — stocks, flows, delays and inertia.



Boundary **EXCEEDED**

124 Mt · **57% polyester**

The microplastics landscape is tightening but rules still target different parts of the problem



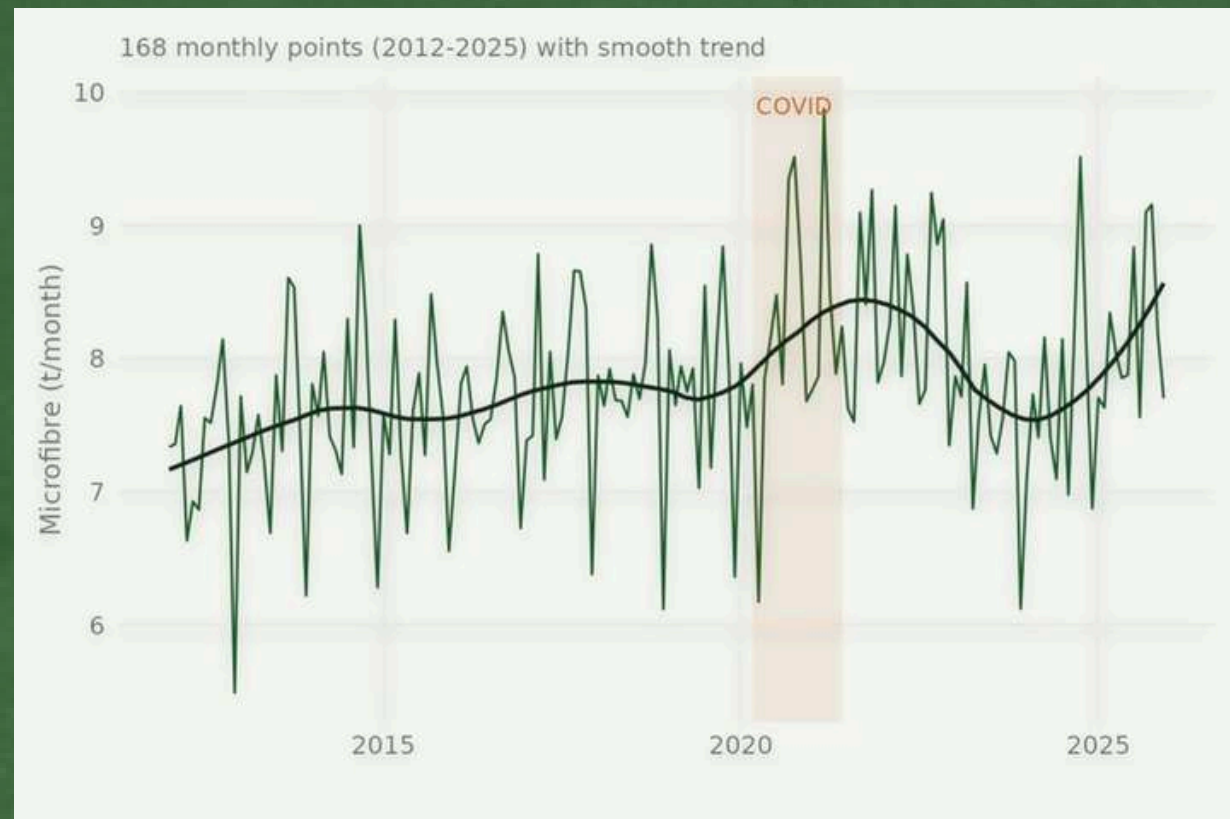
SWISS CONTEXT

FOEN identifies tyre abrasion as a major source and reports that plastic inputs to Swiss soils exceed those to surface waters.

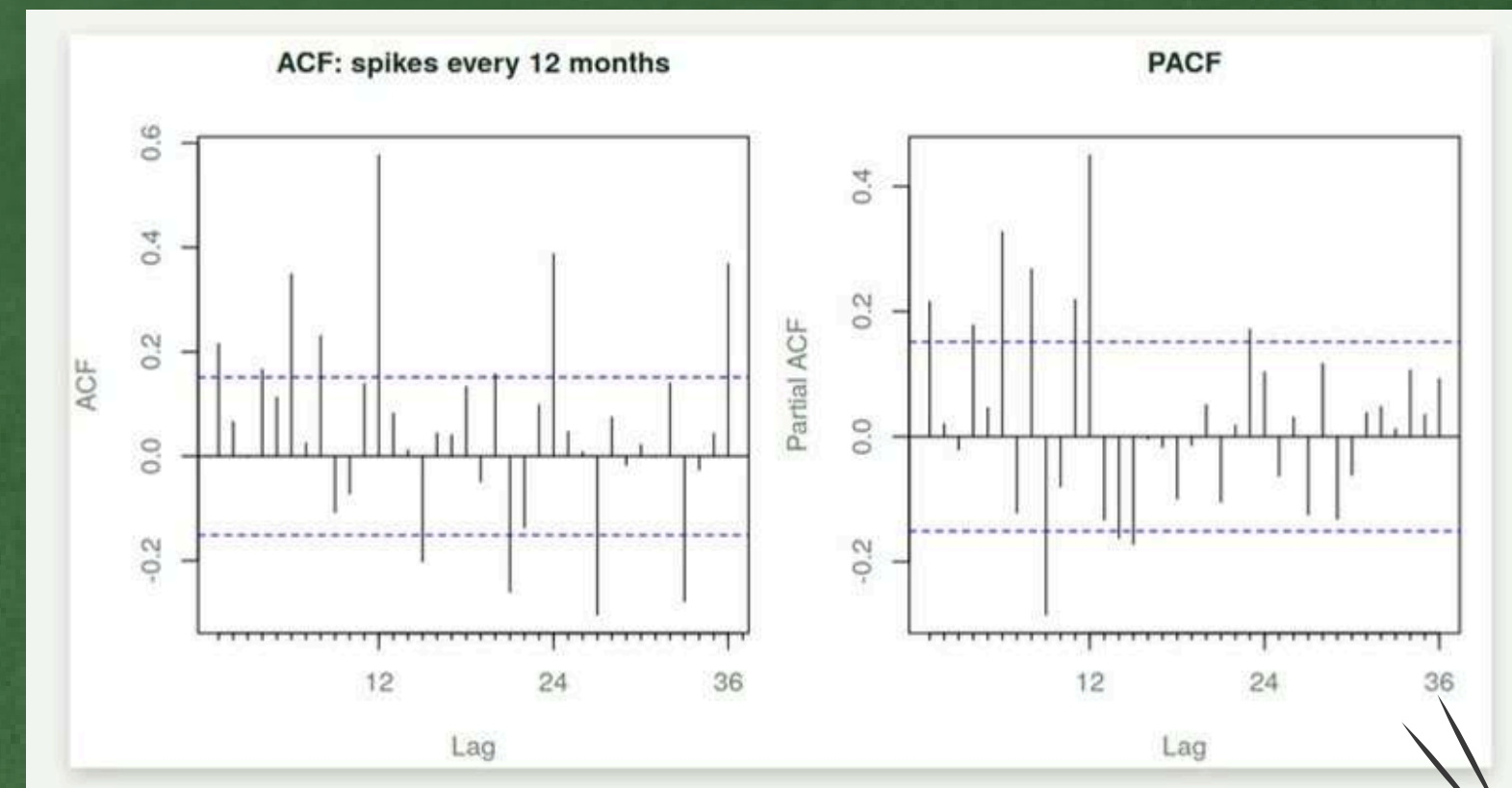


Scope matters: the EU REACH restriction does not cover all secondary microplastics, including wear and fragmentation from finished articles.

Time Series Analysis



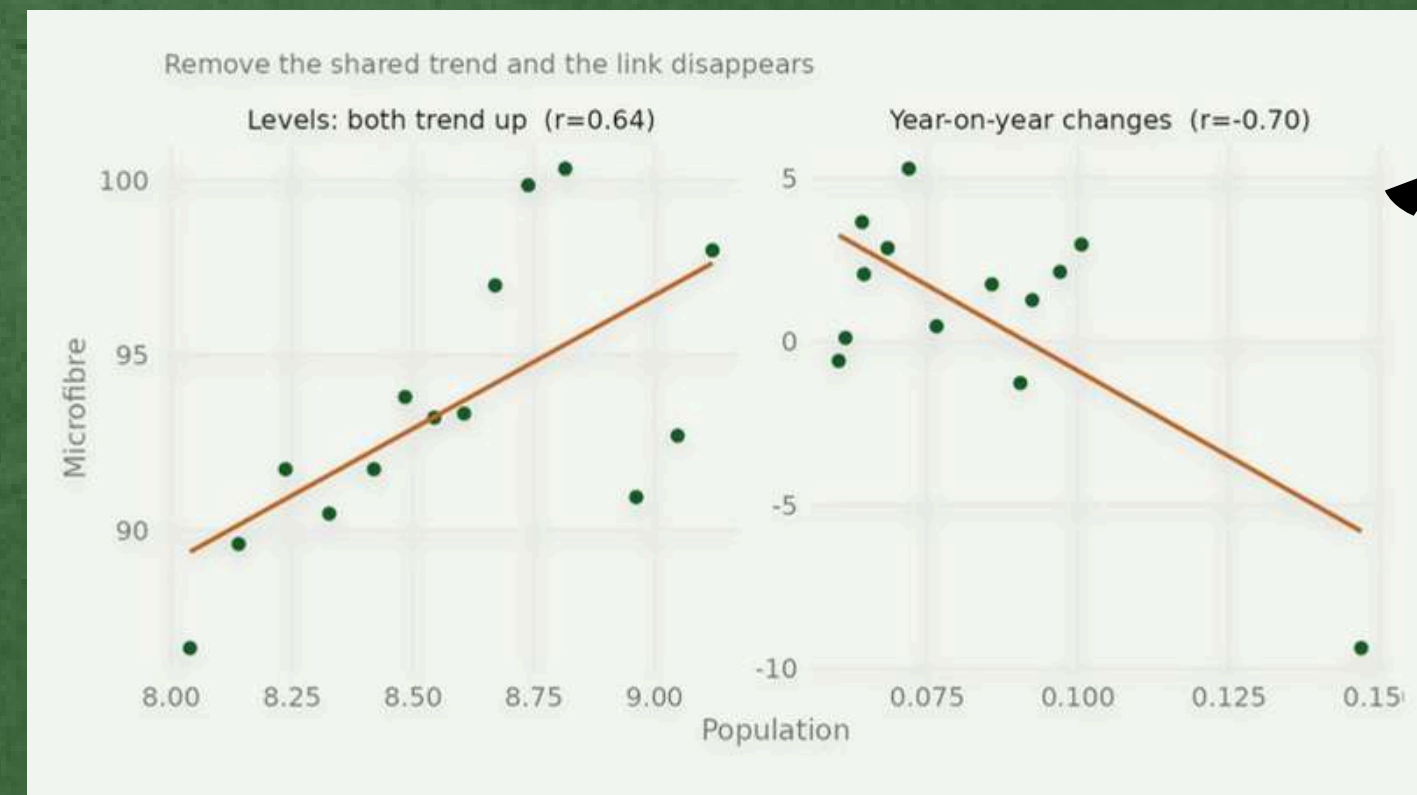
14 years of data reveal a persistent underlying pollution trend.



Pollution follows a predictable 12-month cycle.

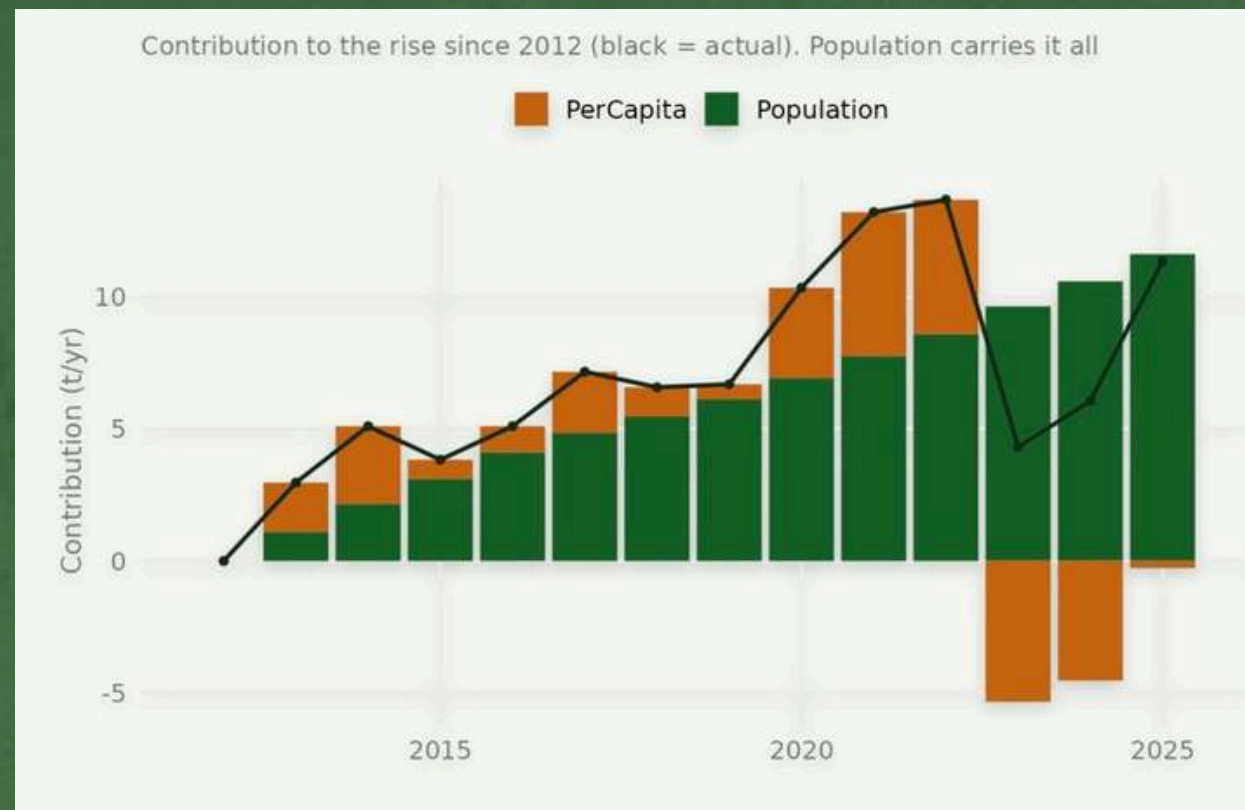


Autumn is the critical intervention window.

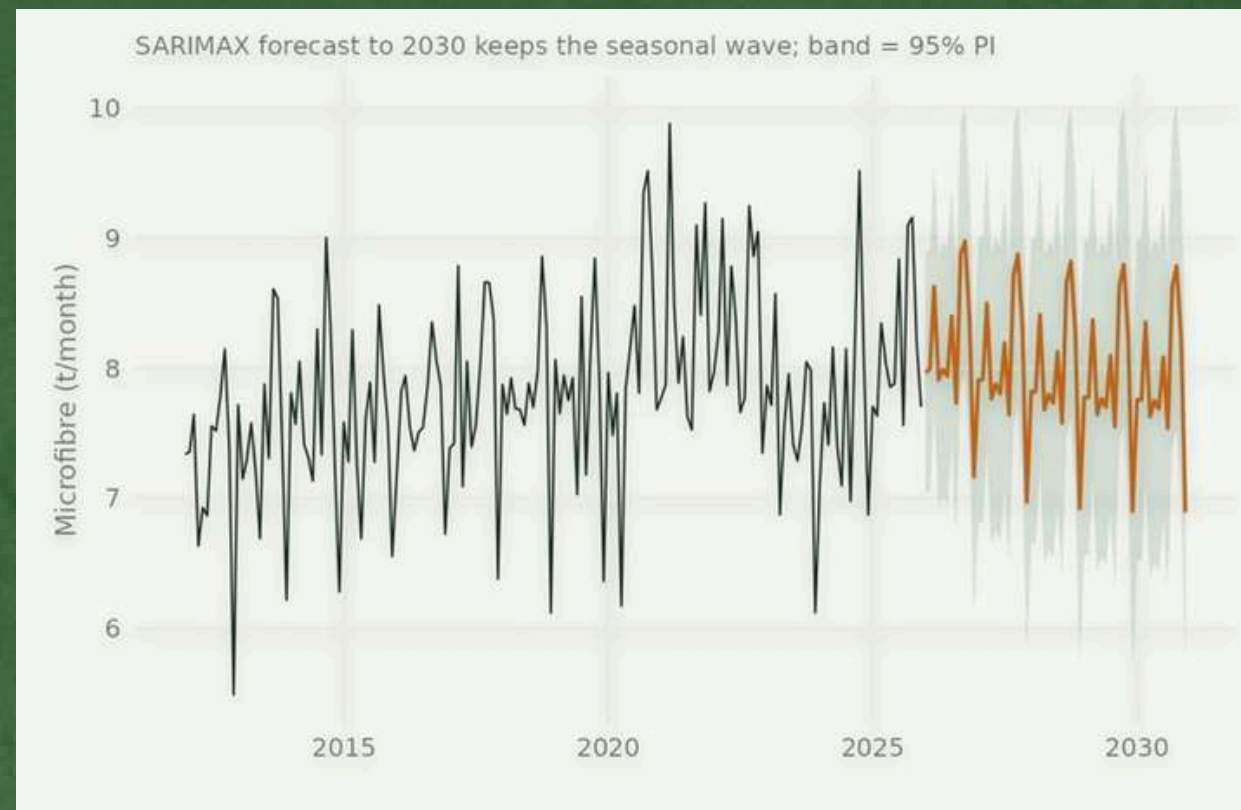


Population explains scale, but not the underlying per-capita trend.

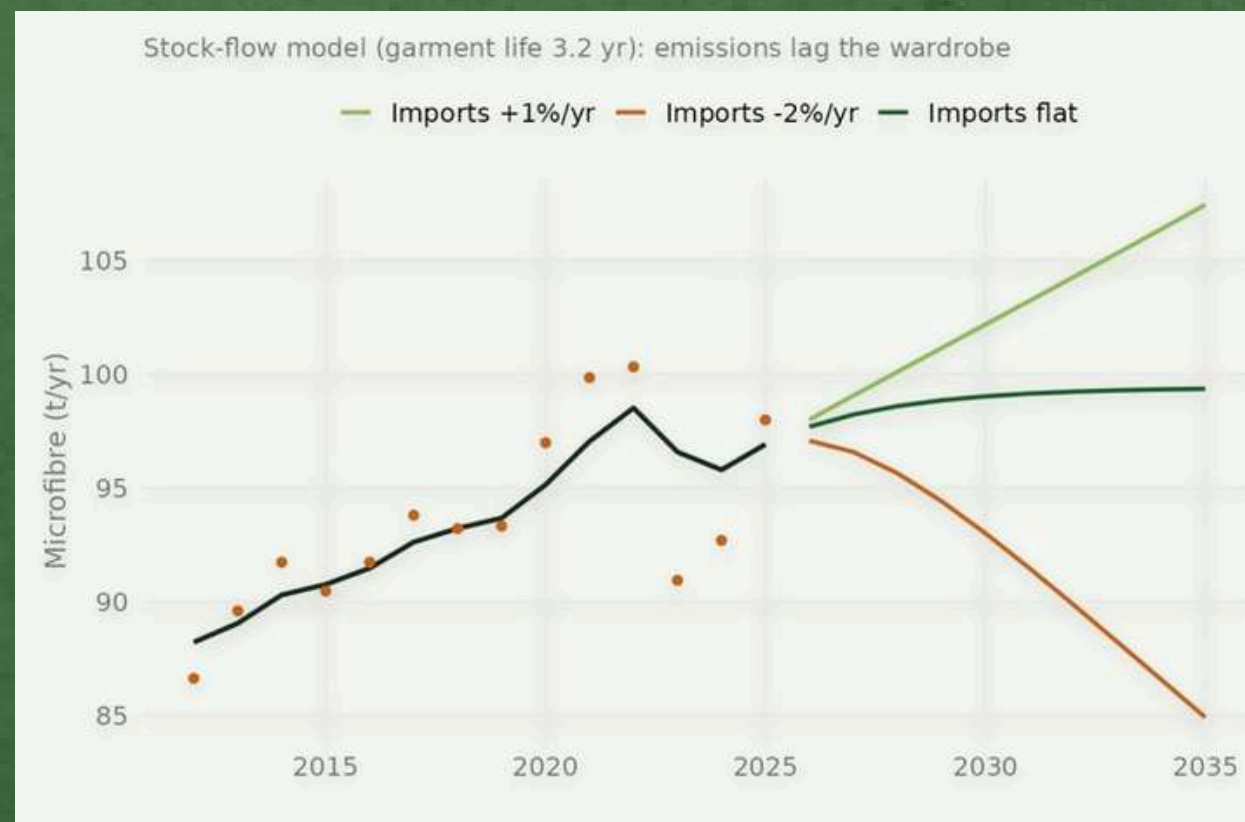
WOW!



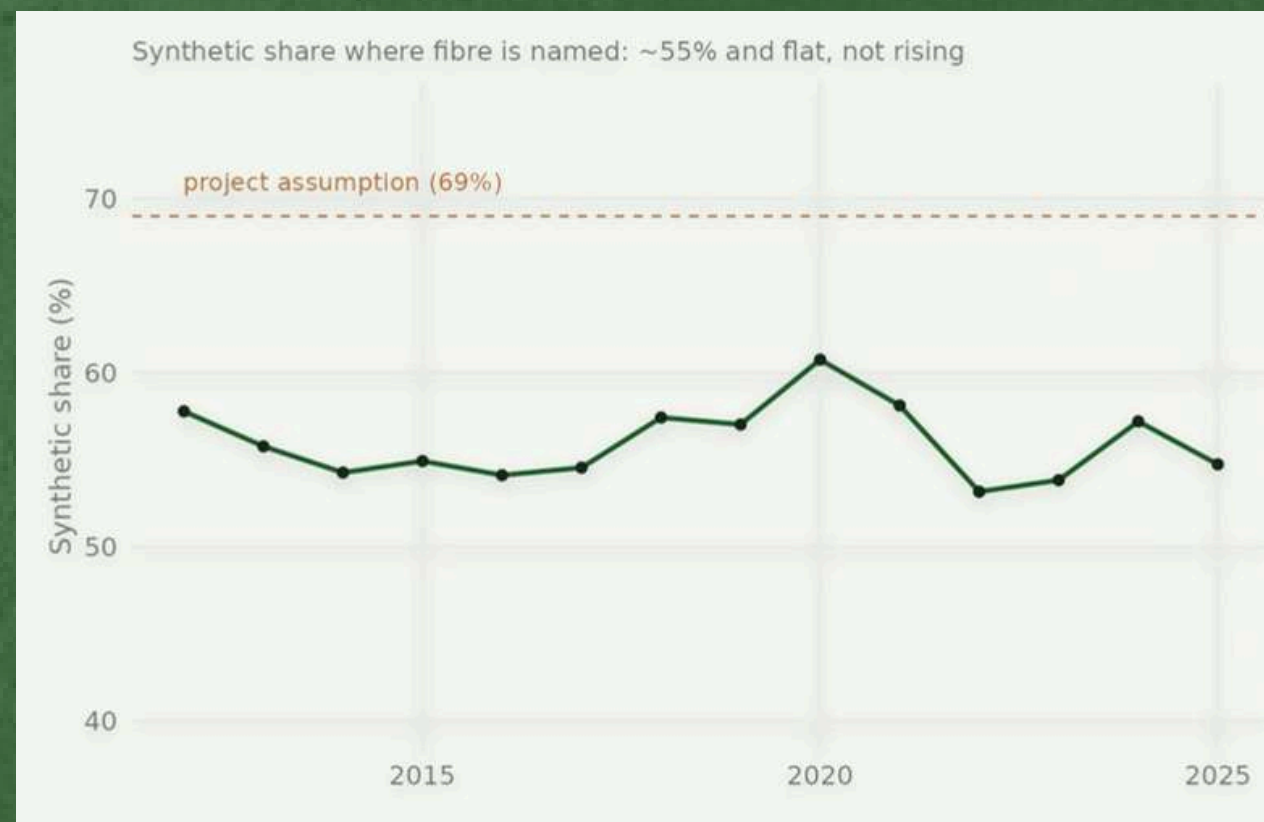
The rise is largely associated with population growth, not higher per-capita consumption.



Without intervention, the seasonal pollution pattern persists to 2030.



Today's pollution comes from years of clothing already in circulation. ~3.2-year garment life creates a delayed response to import changes.



Synthetic share is stable, total volume may be the bigger lever. Below assumed 69%.



What the Time Series Analysis tells us

01.

It's a volume problem, not a behaviour problem. Swiss people aren't buying more per head or switching to synthetics. So "consume less" is the wrong lever; we can't fix this by moralising about consumption.

02.

Make brands pay for how much their clothes pollute. Since the rise comes from synthetic imports, target the source, but charge each garment by how much it sheds, not just its weight. A shedding-based producer fee makes low-shedding, longer-lasting clothes the cheaper choice.

03.

The number is now defensible enough to act on. A backtested monthly forecast (~94 t/yr) means stakeholders can set a target, track it quarter by quarter, and put it in a disclosure, turning the analysis into accountability.



ACTION →
Target Sep–Oct |
Track per-capita
pollution | Address
clothes already in use
| Measure synthetic
volume

Agentic AI Assessment

A verified verdict

1

Certification Relevance Map

Scores 26 frameworks & certifications as RELEVANT or EXCLUDED for this brand with a reason for each.

2

Greenwashing Risk Flags

Catches claims that outrun the evidence, e.g. “eco” on a synthetic-heavy line with no mitigation.

3

Framework Coverage

A 0–100 coverage score for each of SBTi · GRI · MSCI · CSRD.

4

Material-Impact Band

Turns the fibre mix into a microfibre risk band (red / amber / green) using our model.

5

Readiness Score

One 0–100 score across six dimensions, with an at-risk → advanced band.

6

Leverage-Ranked Roadmap

A prioritised fix-list, ordered by what our model proved matters most.

Aligned to leading Standards

1

SBTi - Science based Targets,
near-term and net-zero criteria

2

Global Reporting Initiative, 300
series disclosure

3

MSCI ESG rating - carbon, water,
governance, ethics

4

CSRD / ESRS EU disclosure

GOTS

OEKO-TEX Std100

GRS

RCS

Fair Trade

BCorp

Certification Relevance Map

Greenwashing flag

“Eco” claim on a
synthetic-heavy mix
with no microfibre
mitigation → caught

How the Agent Works



1. User Interaction

- Chat with the agent
- Ask a question
- Request an analysis

2. Agent Orchestration

- Understands intent
- Plans the right tools
- Calls python engines
- Decides what's relevant
- Adapts based on context

3. Analysis Engines

- Certification match
- Greenwashing detection
- Coverage analysis
- Material Impact scoring
- Generate scorecard

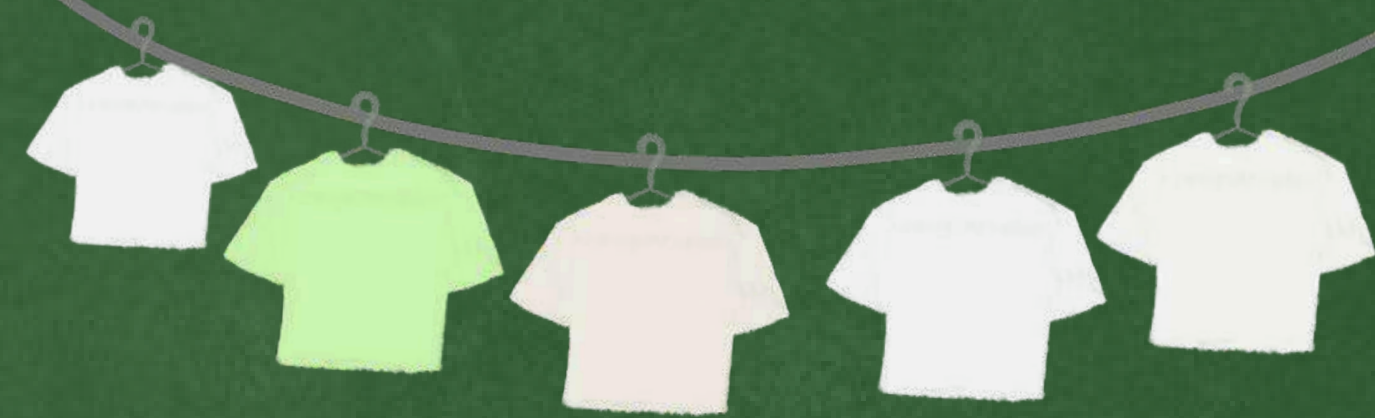
4. Knowledge & Grounding

- 26 Framework catalogue
- Microfibre impact model
- Reference sources

5. Output & Insight

- Readiness Score
- Greenwashing Flags
- Detailed scorecard
- Roadmap

Bridging the gap



1

Assess: 34 frameworks mapped questions

2

Map: Against 26 certifications and standards

3

Score and Flag: Readiness 0-100 + greenwashing risk

4

Prioritise: Fixes ranked by proven leverage

5

Track: Reassess to show progress over time

Who uses it and what they get:

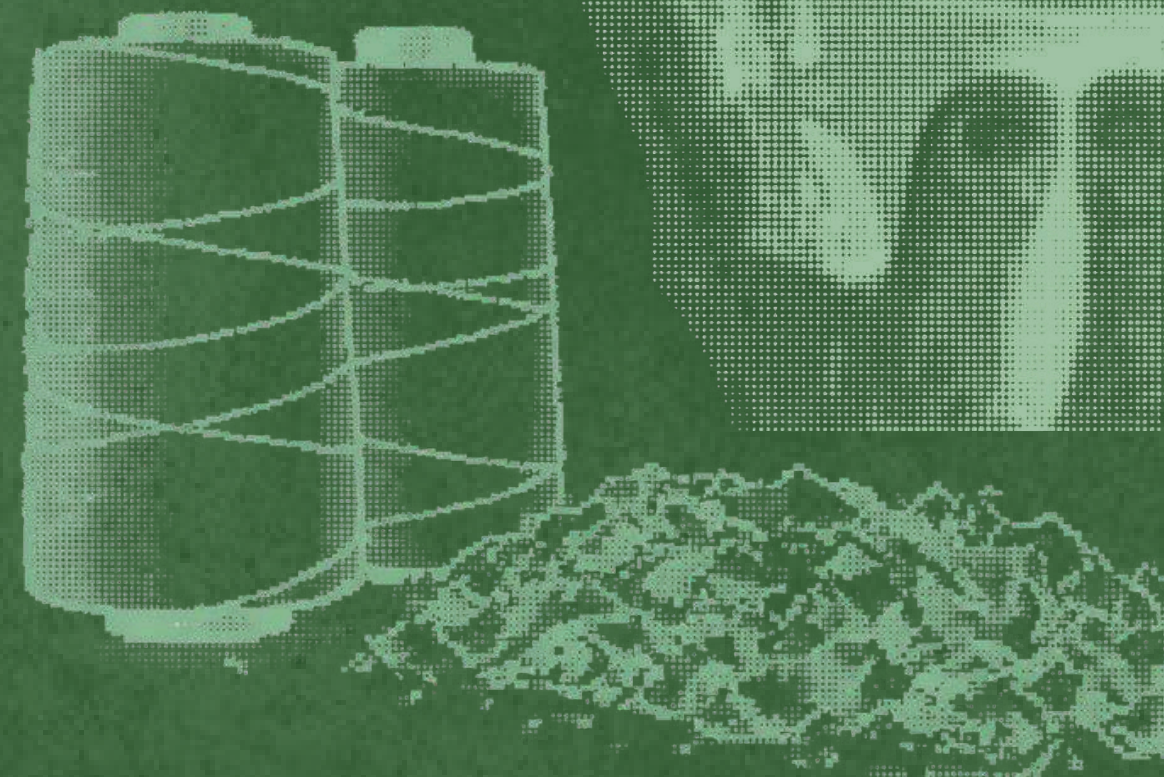
- 1.Brands and sustainability teams - know exactly what to fix first
2. Procurement and sourcing - vet supplier claims first
3. Investors and ESG Analysts - Screen on SBTi, GRI, MSCI, CSRD
- 4.Auditors and CSRD Reporting - an evidence trail for disclosure
- 5.Consultants and certifiers - accelerate readiness prep

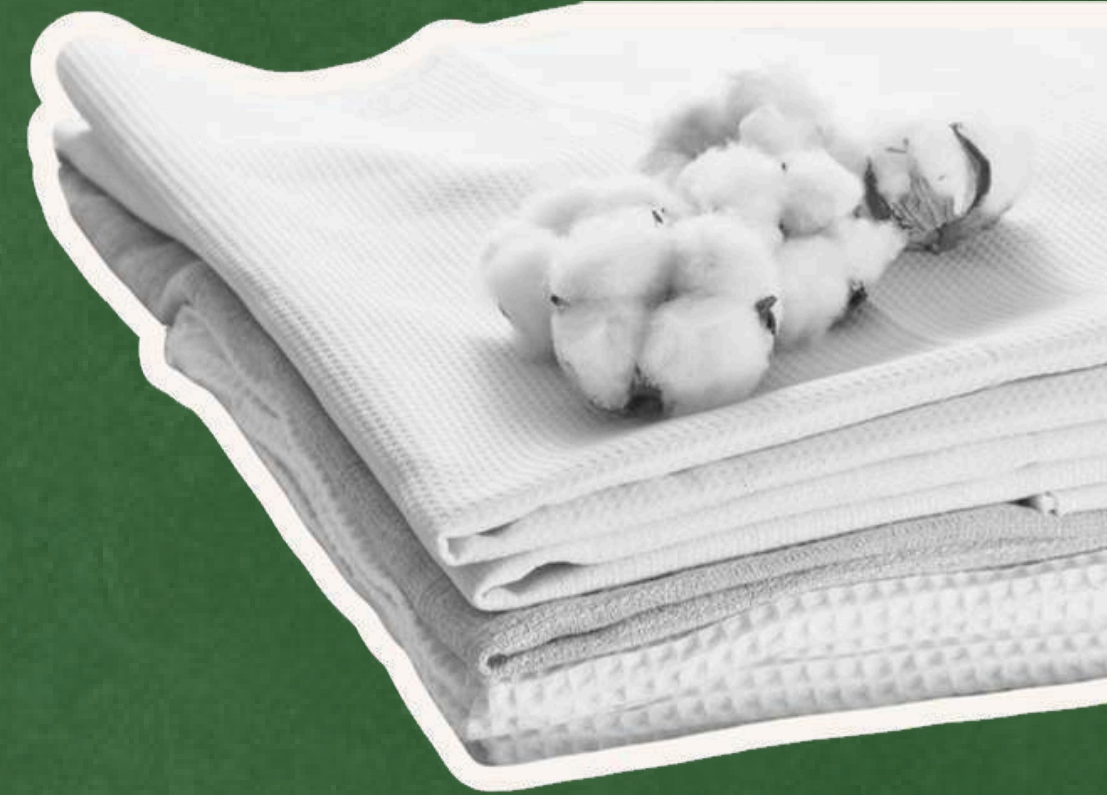
Limitation

The Study was focused on only Microplastics / Synthetic Clothing Data. This study can be extended to other Textile pollutants provided we have enough data and time.

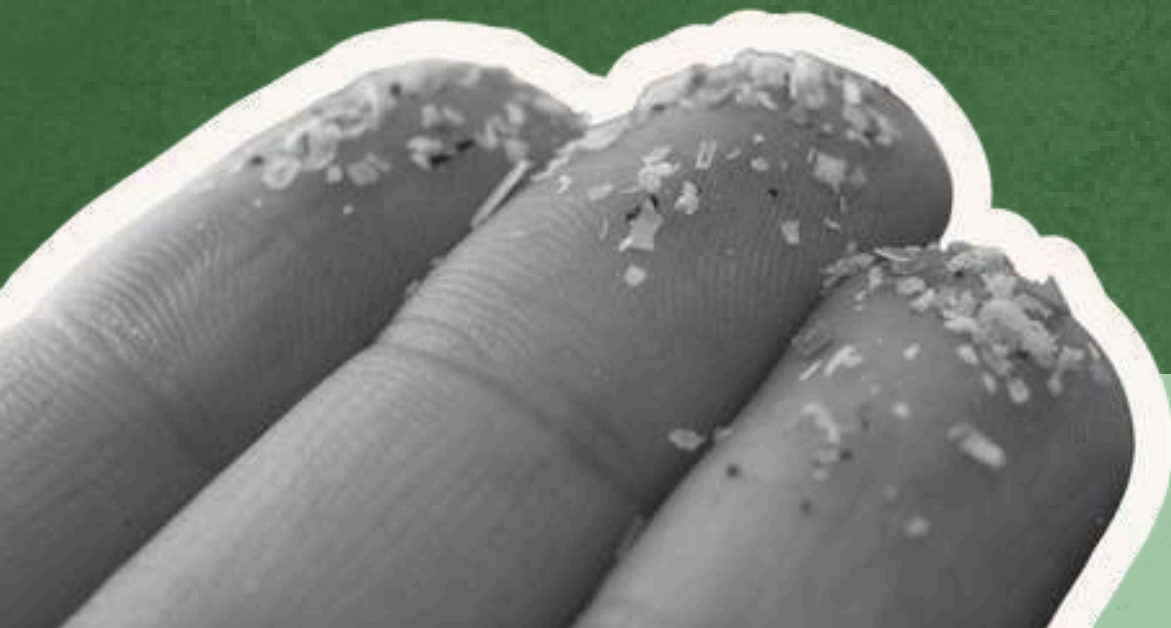
The model shows up to ~90% of the projected microfibre load is avoidable, roughly 520 tonnes over the next decade in Switzerland alone!

We turned an invisible number, **how much plastic our clothes shed**, into a **forecast**, and then into a **tool** that helps brands actually reduce it.





*Thank you
very much!*



We look forward to developing the report next.